VANTAGE
PAYMENT
SYSTEMS

Vantage Payment Systems

Payzone PayWall

Step by step tutorial how to use Payzone PayWall

Payzone®

VANTAGE PAYMENT SYSTEMS

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Payment page

You should have received Payzone_paywall_tutorial.zip

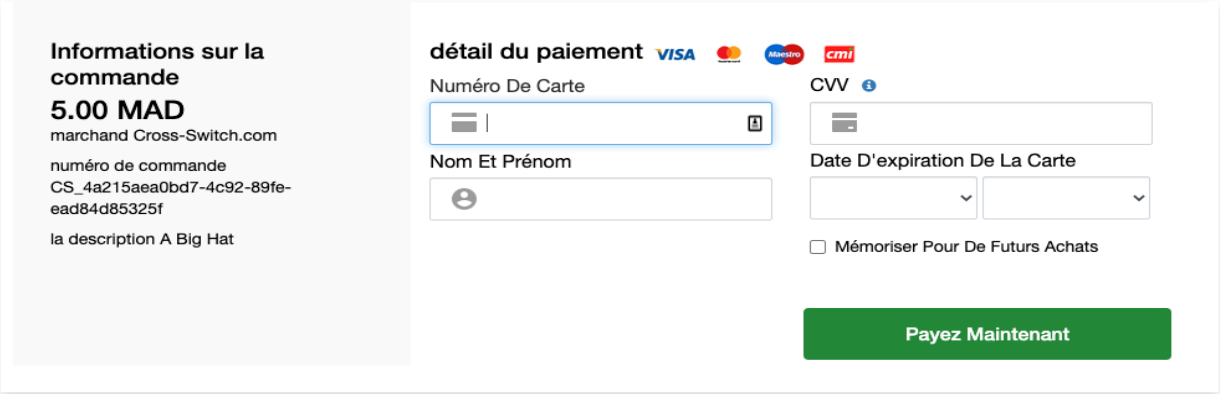
1. Unpack the Tutorial Example Code

Extract the files and place them in a folder on your web server. For this tutorial, we will be using the following:

The files that perform the tasks are as follows:

- **launch_paywall.php** encodes the parameters and sends a POST request to the paywall URL to open it.
- **launch_credentials.inc** contains the URL and secret keys required to open the paywall. The provided file is for demonstration purposes only. Merchant-specific credentials will be provided separately by our team.

The payment page should resemble something like this:



Protection des droits des consommateurs
Par le biais de cette page de paiement, Vantage Payment Systems collecte des données personnelles afin de traiter vos transactions. Ce traitement a été déclaré à la CNDP sous le numéro D-CE-294/2013.
Grâce au protocole de sécurité "Secure Socket Layer" (SSL), vos données sont entièrement cryptées et totalement protégées. Notre certificat SSL a été délivré par la société RapidSSL.

2. Build the Parameter Array

All of the magic happens in **launch_paywall.php** - it JSON encodes the paywall parameters, signs the payload and POST's the results to the paywall URL to invoke the paywall. This tutorial uses PHP, but you can use any language you want, as long as the end result is an HTTP POST to the paywal URL, with the parameters properly JSON encoded and signed.

```
$payload = array(
 // Authentication parameters
 'merchantAccount'      => $merchantAccount,
 'timestamp'       => time(),
 'skin'        => 'vps-1-vue',
 // Customer parameters
 'customerId'      => 'payplus-paywall-poc',    // must be unique for each custumer
 'customerCountry' => 'MA',                     // Fixed value
 'customerLocale' => 'en_US',
 // Charge parameters
 'chargeId'        => time(),      // Optional, if defined, it must be unique for each redirection to the payment page
 'orderId'        => 'order1',      // Optional, to identify the cart
 'price'          => '5',
 'currency'        => 'MAD',
 'description'     => 'A Big Hat',
 // Deep linking
 'mode' => 'DEEP_LINK',                     // Fixed value
 'paymentMethod' => 'CREDIT_CARD',   // Fixed value
 'showPaymentProfiles' => false,
  'callbackUrl' => 'https://test-merchant.ma/callback',
  'successUrl' => "https://test-merchant.ma/success",
  'failureUrl' => "https://test-merchant.ma/failure",
  'cancelUrl' => "https://test-merchant.ma/cancel",
)
```

The first thing **launch_paywall.php** does is create an array of parameters. A typical case might look like this (replace the hard-coded values with appropriate ones, of course).

Refer to the reference section "Input Parameters" for additional launch options.

3. Encoding the Parameters

The next step is to sign and encode the parameters. The payload is signed using SHA256 and the paywall secret key.

```
$json_payload = json_encode($payload);
$signature    = hash('sha256', $paywallSecretKey . $json_payload);
```

4. POST the encoded payload and signature to the paywall URL

Finally, you POST the encoded parameters to your paywall URL to invoke the paywall:

```
<form id="openPaywall" action="<?php echo $paywallUrl; ?>" method="POST">
    <input type="hidden" name="payload" value='<?php echo $json_payload; ?>' />
    <input type="hidden" name="signature" value="<?php echo $signature; ?>" />
</form>
```

Note that **$merchantAccount**, **$paywallSecretKey** and **$paywallUrl** are assigned by Payzone.

For the tutorial, we have copied these values into **launch_credentials.inc**.

5. Handling Real-Time Notifications

In addition to the postMessage events which you can use to manage your display to the customer, Payzone sends a real-time notification to your server any time a charge changes state. Please provide notification Endpoint on your side for configuration. You should use these notifications to fulfill goods to your customers, and also to be notified of refunds, chargebacks or asynchronous completion events.

6. Setup the Notification URL

There are two ways to specify the notification URL for a transaction.

- Setup the global notification in your merchant account
- Send in the notification URL in paywall payload

If you want to set up the notification URL in your merchant account, please contact your account manager.
The notification URL via payload has higher priority than the then on in your merchant account. That is, if you pass in the notification URL via the payload, the notification is sent to it instead of the URL you have set via your merchant account.

7. Validate the Notification

- Retrieve the content of the callback
- Calculate the signature by calculating an HMAC hash on the content and your notification key using sha256 encoding
- Retrieve the ***X-callback-signature*** header
- Compare the signature with your calculated signature

Here is an example in PHP:

```
$input = file_get_contents('php://input');
$signature = hash_hmac('sha256',$input,$notificationKey);
$headers = apache_request_headers();
if (strcasecmp($signature, $headers['X-callback-signature']) == 0 )
{
  // Process the notification here
}
```

8. Notification Resent

After successfully processing the notification, please return HTTP status 200 to CS. Whenever Payzone sends a notification to you, if you do not respond with a 200 OK, then it assumes that you did not receive it and the notification is placed in a queue. It will be resent to you periodically over the next 3 days until you either respond with a 200, or the time limit expires.

9. Sample notification:


```
{
   "id": "202110112305",
   "orderId": "order 1",
   "internalId": "2347897968234687234",
   "status": "CHARGED",
   "merchantAccount": "Payzone",
   "lineItem": {
       "amount": 12.7,
       "currency": "MAD",
       "description": "5 Golden Rings",
       "taxCode": "NT",
       "taxIncludedInPrice": true
   },
   "transactions": [
       {
           "timestamp": "2021-10-11T15:07:17Z",
           "gatewayProvidedId": "dd317ae1-f9cb-47ef-8a8b-10e8a754c5a2",
           "type": "CHARGE",
           "state": "APPROVED",
           "amount": 12.7,
           "currency": "MAD",
           "resultCode": 0,
           "responseText": "Success"
       }
   ],
   "paymentType": "CREDIT_CARD",
   "paymentMethod": "Visa",
   "created": "2021-10-11T15:07:16Z",
   "amountRemaining": 12.7
}
```


Refer to the reference section “Notification Parameters” for additional options.

10. Reference

- Input Parameters

The paywall supports a number of parameters in addition to those used in this tutorial. To send additional parameters, simply include them in the $payload array prior to JSON encoding.

Parameter	Required	Description
Authorization Parameters		
merchantAccount	Yes	Your merchant Account Name, provided by Payzone.
timestamp	Yes	The current time measured in the number of seconds since the Unix Epoch (January 1 1970 00:00:00 GMT). This is used to calculate the signature, and is only valid fo 30 minutes, limiting exposure to replay attacks.
Charge Parameters		
chargeId	No	A unique value indicating what was purchased, usually your internal transaction ID. It will be passed back to you in the notifications so that you can fulfill the purchase. If you do not provide a chargeId, Payzone will create one for you.
orderId	No	It is an identifier that distinguishes the cart or the order.
price	Yes	The price is the amount to charge the customer
currency	Yes	the ISO 4217 currency code for the transaction currency (MAD,USD, EUR, etc)
description	Yes	A description of what the customer is buying. This will appear on the UI, on the purchase receipt and in transaction reports.
memo	No	A string that will be recoded with the charge, and included in the memo column of the transaction reports
chargeProperties	No	Optionally adds properties to the charge. Properties are name/value pairs in this format:

		"{"name1":"value1","name2":"value2"}". '=' and ' ' are not allowed in the name. The total size of the string cannot exceed 3500 characters.
Customer Parameters		
customerId	Yes	your ID for this customer. This is used for fraud checks and reporting, and is returned to you so that you can fulfill the purchase.
customerCountry	No	A two-letter ISO country code. If it is required but not provided, the paywall will use IP Geo-location or ask the customer to select their country.
customerName	No	The full name of the customer.
customerEmail	No	The Email address where the customer will receive confirmation emails, if Payzone is sending confirmation emails. If it is required and not provided, the paywall will ask the customer to provide it.
customerStateProvince	No	The state or province of the customer who will be charged. It is needed to calculate tax in some countries. If it is needed and not provided, the paywall will ask the customer to provide it.
customerCity	No	The name of the city the customer lives in.
customerPostalCode	No	The postal code of the customer who will be charged. It is needed for some tax calculations in the US. If it is needed and not provided, the paywall will ask the customer to provide it.
customerStreetAddress	No	The street address of the customer, which may be required to calculate tax on physical goods. If it is required and has not been supplied, the paywall will ask the customer to provide it.

customerStreetAddress2	No	The street address of the customer, which may be required to calculate tax on physical goods. If it is required and has not been supplied, the paywall will ask the customer to provide it.
customerDOB	No	The date of birth for the user, in the format YYYY-MM-DD. If provided it is used for fraud calculations and can be useful when resolving customer complaints. Some payment providers may require a DOB if your product is intended for children.
customerLocale	No	Used to override the default language of the user's country. This is a 2-letter ISO language code and 2-letter ISO country code separated by underscore Supported locales: en_US, fr_FR, pt_BR, ar_EG, da_DK, de_DE, es_ES, fi_FI, it_IT, ja_JP, ko_KR, nl_NL, no_NO, pl_PL, ru_RU, sv_SE, tr_TR
Paywall Options		
successUrl	No	The URL the paywall will redirect to after a successful purchase.
failureUrl	No	The URL the paywall will redirect to after a failed purchase attempt, either due to a fraud rejection, inability to communicate with the PSP or a rejection from the PSP.
cancelUrl	No	The URL the paywall will redirect to if the customer cancels the purchase flow.
flowCompletionUrl	No	The URL the paywall will redirect to at the end of the purchase flow for any result if the successUrl, failureUrl or cancelUrl was not specified.
callbackUrl	No	The URL at which you will receive real-time notifications any time a charge changes state.

		Alternatively, it can be setup globally for all all transactions by Payzone support.
cssUrl	No	An HTTPS URL to custom CSS that will be applied to the paywall when it is opened in a laptop or desktop browser
mobileCssUrl	No	An HTTPS URL to custom CSS that will be applied to the paywall when it is opened on a mobile device
savePaymentProfile	No	Boolean flag, when set to 'true' payment information will automatically be save for offsite methods where we cannot show the opt-out checkbox.
showPaymentProfiles	No	Boolean flag, when set to 'true' Paywall will display previously created stored payment profiles along with new credit cards option.
'Deep Link' mode parameters		
mode	Yes	Set to 'DEEP_LINK' to use deep linking.
paymentMethod	Yes	Should be set to “CREDIT_CARD”.

- Notification parameters:

name	type	required
status	*string*	*required*
Describes the status of the charge. Valid values: • AUTHORIZE_PENDING a request was made to capture an amount with the AUTHORIZE command. If present the merchant must process the charge.paymentOption to advance the charge status; in other cases the merchant		

just needs to wait until the next real-time notification indicating the processing is complete.

- AUTHORIZED the amount has been approved and can be captured by the merchant with the SETTLE command or the funds can be released with the AUTH_REVERSAL command.
- AUTH_REVERSED the amount is no longer available for capture. This is a final status.
- CANCELLED the amount is not available because the customer cancelled the charge. Not all processors and methods support a CANCELLED status and for these the charge would end up as DECLINED or ERROR. This is a final status.
- DECLINED the amount is not available because the processor rejected the charge. This is a final status.
- CHARGE_PENDING a charge has been initiated and the request is awaiting a response from the PSP.
- CHARGED the amount has been captured and the purchased product can be released. Note that partially refunded charges are in CHARGED status until they are fully refunded.
- CHARGED_BACK the customer has reclaimed the captured amount and it is no longer available.
- CHARGEBACK_REVERSED the amount has been disputed and recaptured after previously having been CHARGED_BACK.
- CHARGEBACK_PENDING a chargeback has been initiated by the customer but the funds have not been returned
- REFUND_PENDING a refund has been initiated (typically by the merchant) but the funds have not been returned to the customer yet. The merchant just needs to wait until the next real-time notification indicating the processing is complete.
- REFUNDED a full refund has been initiated (typically by the merchant) and the funds have been returned to the customer. Note that a partial refund will remain in the CHARGED status. This is a final status.
- ERROR the amount is not available for any reason. For details consider looking at the charge history in the transactions[] section. This is a final status.
- CREDITED_PENDING the credit is in progress.
- CREDITED the amount is credited to customer.

paymentOption	*PaymentOption (see fields)*
Details for *paymentOption*	

The *paymentOption* is returned in the charge whenever additional customer interaction is needed to complete the purchase. Typically, this is used for 3D-Secure and online banking verification.

Important the *paymentOption* object is transient and not persisted with the charge; therefore, it must be processed by the merchant directly from the POST charges response body. It will not be available in subsequent GET charges or real-time notifications.

paymentProfile	*PaymentProfileStaus (see fields)*

Details for *paymentProfile*

For some purchases a new stored payment profile is created as part of the charge, for example, when processing an INITIAL recurring purchase. For these cases the details about the new payment profile are returned in a *paymentProfile* object within the charge.

Important in some cases the *paymentProfile* will not be returned in the initial POST charges response; instead, the section will be added to the charge later and made available to the merchant using a real-time notification callback.

command	string	required

Details for *command*

The action most recently performed on charge request.

allowedCommands	array	required

Details for *allowedCommands*

List of commands that can be applied to this charge based on its current status.

transactions	array	required

Details for *transactions*

Historical details about this charge

type	String *(see details)*	required
Details for *type* API version. Always v3_2 when ChargeFormatV3_2 is configured. • Type Values: v3_0, v3_1, v3_2		
created	datetime	required
Details for *created* Time this charge was first submitted in ISO-8601 format		
lastModified	datetime	required
Details for *lastModified* Time this charge was last modified in ISO-8601 format		
merchantAccount	string	required
Details for *merchantAccount* Name of the merchant account		
LGID	integer	required
Details for *LGID* Unique transaction ID generated by the payment system		

taxAmountCorrected	boolean

Details for *taxAmountCorrected*

This flag indicates whether the tax percentage/amount given in the request is corrected or not. This will appear in the response only if tax amount is corrected and is applicable only for India as of now.

lineItemOriginal	LineItem *(see fields)*

Details for *lineItemOriginal*

The line item original object records the original incoming tax details if the tax amounts/percents are corrected based on the tax calculation on EP side.This will appear in the response only if tax amount is corrected and is currently applicable only to India.

taxInvoices	array of InvoiceResponse *(see fields)*

Details for *taxInvoices*

This is an array of tax invoices stored as part of a specific charge.

id	string of length 0 to 63

Details for *id*

Merchant provided id to uniquely identify the transaction. If not provided, EP system will generate a unique ID and return that in the response. Recommended that merchant passes unique id in this field.

memo	string of length 0 to 255

Details for *memo*

Metadata the merchant can store with the charge. The memo will be passed back with real-time notifications and included in transaction and settlement reports. Recommended that merchant passes business data in this field.		
properties		object
Details for *properties* Properties are sent as name/value pairs in this format: "{"name1":"value1","name2":"value2"}" Property names are case sensitive, so 'Name1' is a different property than 'name1'.		
lineItemProperties		object
Details for *lineItemProperties* Additional line item properties are sent as name/value pairs in a list in this format: "{"lineItemProperties": [{ "quantity": "1","description": "Product 1 description", "deliveryMethod": "Included" },{ "quantity": "3", "description": "Product 2 description", "deliveryMethod": "Setup from My Account" }]}" Property names are case sensitive, so 'quantity' is a different property than 'Quantity'.		
recurrence		object *(see fields)*
Details for *recurrence* Recurrence fields are used for recurring transactions. The initial charge must be done with the customer present and CVV must be supplied for credit cards. The renewal is done subsequently without the customer present and CVV is not required for credit cards.		
idempotencyId		string of length 0 to 63
Details for *idempotencyId* If specified guarantees this same charge has not already been processed. This could be used if client wants to timeout and retry without having to check status of previous charge submission. If first charge attempt was received, the retry will fail.		
lineItem	LineItem *(see fields)*	required

Details for *lineItem* Details about the item being purchased.		
customer	Customer *(see fields)*	required
Details for *customer* The customer object keeps details such as the tax and shipping address and fraud blocks related to an individual. When supplied with a charge the customer record is updated.		
payment	Payment *(see fields)*	required
Details for *payment* The payment object keeps details such as the billing address and account number related to a specific payment method. When supplied with a charge the customer record is typically not updated with the billing address. To update the customer record, include the new address in the customer object. For credit card payments, it is possible to encrypt the card number in the browser and send it encrypted all the way to our backend payment processing servers. This feature is known as client-side encryption and may greatly reduce a merchant's risk and PCI concerns. Passing a hash of the public key returned by /api/v3/encryptionkey API in the header of the request indicates that the credit card number is encrypted.		
metadata	Metadata *(see fields)*	
Details for *metadata* The metadata object keeps optional merchant and processor specific information that might be needed to complete the payment.		
paymentOption fields		
paymentServiceURL	string	

Details for *paymentServiceURL*

If present then the customer's browser should be redirected to this URL to complete the purchase. In most cases a simple redirect (HTTP GET) is sufficient; however, for some payment methods the merchant will need to adjust the behavior based on other fields such as offsiteMethod and uiType.

For typical use cases, when the verification steps are complete the customer's browser will be redirected back to the web page that the merchant specified in the charge.metadata.returnURL field when the charge was first created with POST charges.

Note that depending on the flow the customer might not ever be redirected back to the merchant returnURL. For example, if the customer fails the off site verification the browser is typically redirected to the merchant failureURL. In another example the customer might close the browser and abandon the flow. For these reasons the merchant should always rely on the real-time notifications as the ultimate source of truth when evaluating if a charge was successful.

uiType	string *(see details)*

Details for *uiType*

This describes the constraints the off site web page. The merchant should take these constraints into account when deciding how to redirect the customer's browser to the paymentServiceURL.

- IFRAME and PRINT_IFRAME indicate the off site has no constraints and should render properly regardless of if it is displayed in an IRAME, a pop-up browser, or replaces the merchant content.
- POPUP_WINDOW and PRINT indicate the off site will not render correctly in an IFRAME; the merchant must either open the URL in a new pop-up browser window or must replace the merchant content so the URL opens in the top of the current frame.
- PRINT and PRINT_IFRAME indicate the off site has no "return to merchant" UI element, for example when paymentServiceURL refers to a PDF file.
- Type Values: IFRAME, POPUP_WINDOW, PRINT, PRINT_IFRAME

windowWidth	integer

Details for *windowWidth* When specified this gives the minimum browser window content area width in pixels to provide the best customer experience for an off site flow.		
windowHeight		integer
Details for *windowHeight* When specified this gives the minimum browser window content area height in pixels to provide the best customer experience for an off site flow.		
offsiteMethod	String *(see details)*	required
Details for *offsiteMethod* *With very few exceptions, the merchant can expect this value to be GET for most payment methods.* Determines if the paymentServiceURL should be redirected to with a HTTP GET or a POST. • Type Values: GET, POST		
postForm		object
Details for *postForm* *With very few exceptions, the merchant can expect there to be no postForm for most payment methods.* When offsiteMethod is POST this field may contain a set of string value pairs in this format "{"name1":"value1","name2":"value2"}". If present then these should be used to build an HTTP FORM POST. Example form built to redirect the customer's browser to an off site with a form post		

```
<form id="auto-post" action="paymentServiceUrl" method="offsiteMethod">
  <input type="hidden" name="name1" value="value1">
  <input type="hidden" name="name2" value="value2">
</form>

<script>
  document.getElementById("auto-post").submit()
</script>
```

paymentProfile fields

id	string	required

Details for *id*

The ID of the newly created payment profile. Supply this value for subsequent purchases as charge.payment.storedPaymentProfileId.

paymentMethod	string	required

Details for *paymentMethod*

This typically mirrors the value supplied as charge.payment.paymentMethod; however, in some cases this may be a more specialized value such as VISA in lieu of CREDIT_CARD. Typical values are VISA, MASTERCARD, CASH, etc.

paymentMethodType	string *(see details)*	required

Details for *paymentMethodType*

This mirrors the value returned in charge.payment.paymentType. The type of payment method used.

- Type Values: CREDIT_CARD, DEBIT_CARD, PREPAID_CARD, BANK_TRANSFER, VIRTUAL_ACCOUNT, DIRECT_DEBIT, CASH, CHECK, OFFER, WALLET, CARRIER_BILLING

type	string *(see details)*

Details for *type*

This mirrors the value supplied in charge.recurring.type. This field is omitted for non-recurring charges.

- Type Values: INITIAL, RENEWAL

state	String *(see details)*	required

Details for *state*

Indicates if the stored payment profile is ready to be used for a new charge.

- FAILED failed to create payment profile.
- UNVERIFIED payment profile was created successfully, but unverified OR pending on customer action. No success transaction completed with this recurring.
- VERIFIED payment profile was created successfully, and verified because there is success transaction completed with this payment profile.
- EXPIRED The payment profile is expired or removed.
- Type Values: FAILED, UNVERIFIED, VERIFIED, EXPIRED

reserved fields		
purchaseFlowVersion	string	required

Details for *purchaseFlowVersion*

For EP internal use

lineItemOriginal fields		
currency	string of length 3 to 3	required
Details for *currency* ISO 4217 Three letter currency code code for the transaction currency (MAD,USD, EUR, etc)		
amount	number	required
Details for *amount* The price of the goods being sold. The customer will be charged this amount, plus any taxes that are not included in the price.		
description	string of length 0 to 256	required
quantity	number	
Details for *quantity* The quantity of the items.		
deliveryMethod	string	
Details for *deliveryMethod* The delivery method of the items.		
taxCode	string of length 2 to 64	
Details for *taxCode*		

One of the supported tax codes. If none is specified then the one configured as default by your EP TAM will be used.

- DG (Digital Good)
- VC (Virtual Currency)
- NT (Non-Taxable)
- NA (Unknown)
- PG (Physical Goods)
- SV (Service)
- DG_INTERNET_ADS
- DG_ONLINE_CONTENT_SERVICE
- DG_LIC_SERVICE_SOFTWARE_DATABASE

taxAmount	number
Details for *taxAmount* Tax amount for this transaction.	
taxPercent	number [0...1]
Details for *taxPercent* Tax rate for this charge.	
taxProvider	string *(see details)*
Details for *taxProvider* Indicates who calculated tax. Values include 'MERCHANT', 'ADAPTER', 'NONE' • Type Values : MERCHANT, ADAPTER, NONE	
taxIncludedInPrice	boolean

Details for *taxIncludedInPrice*

True if tax is included in the amount. If this field is not specified then then the value configured by your EP TAM will be used.

baseAmount	number

Details for *baseAmount*

Transaction amount excluding taxes. Currently sent only for India and Argentina

taxDetail	object

Details for *taxDetail*

Individual tax percents and their associated tax amounts can be sent as key value pairs.It is currently sent as part of the response only for India to specify the tax split ups CGST,SGST,IGST.

Example for GST India:

```
"taxDetail": {
   "cgstPercent": "0.09",
   "sgstPercent": "0.09",
   "igstPercent": "0.00",
   "cgstAmount": "18",
   "sgstAmount": "18",
   "igstAmount": "0.00"
 }
```

recurrence fields

type	string *(see details)*	required

Details for *type*

INITIAL is used to signify first transaction and subsequent recurring transactions should have type as RENEWAL.

- Type Values: INITIAL, RENEWAL

lineItem fields		
currency	string of length 3 to 3	required
Details for *currency* ISO 4217 Three letter currency code code for the transaction currency (MAD,USD, EUR, etc)		
amount	number	required
Details for *amount* The price of the goods being sold. The customer will be charged this amount, plus any taxes that are not included in the price.		
description	string of length 0 to 256	required
Details for *description* Description of the items.		
quantity	number	
Details for *quantity* The quantity of the items.		
deliveryMethod	string	
Details for *deliveryMethod* The delivery method of the items.		

taxCode	string of length 2 to 64

Details for *taxCode*

One of the supported tax codes. If none is specified then the one configured as default by your EP TAM will be used.

- DG (Digital Good)
- VC (Virtual Currency)
- NT (Non Taxable)
- NA (Unknown)
- PG (Physical Goods)
- SV (Service)
- DG_INTERNET_ADS
- DG_ONLINE_CONTENT_SERVICE
- DG_LIC_SERVICE_SOFTWARE_DATABASE

taxAmount	number

Details for *taxAmount*

Tax amount for this transaction.

taxPercent	number [0...1]

Details for *taxPercent*

Tax rate for this charge.

taxProvider	string *(see details)*

Details for *taxProvider*

Indicates who calculated tax. Values include 'MERCHANT', 'ADAPTER', 'NONE'

- Type Values: MERCHANT, ADAPTER, NONE

taxIncludedInPrice		boolean
Details for *taxIncludedInPrice* True if tax is included in the amount. If this field is not specified then then the value configured by your EP TAM will be used.		
baseAmount		number
Details for *baseAmount* Transaction amount excluding taxes. Currently sent only for India and Argentina		
customer fields		
id	string of length 0 to 63	required
Details for *id* Unique id to identify the customer of merchant. If no matching customer record is found, a new customer record will be created. If matching Id is found in EP system, the charge will be associated with that customer.		
firstName		string of length 0 to 50
lastName		string of length 0 to 50
companyName		string of length 0 to 50
email		string of length 0 to 100
streetAddress		string of length 0 to 100
streetAddress2		string of length 0 to 20
city		string of length 0 to 30

stateProvince	string of length 0 to 20
postalCode	string of length 0 to 20
countryCode	string of length 2 to 2 (ISO 3166-1 alpha-2 customer country code.)
ipAddress	string of length 0 to 16
customerType	string *(see details)*

Details for *customerType*

Identifies the type of customer to specify individual/business customers.This field is applicable only for India as of now. The supported customer types are

- B2B (Business customers)
- B2C (Individual customers)
- BOTH (Supports both Business and individual customers)
- Type Values: B2B, B2C, BOTH

document	string

Details for *document*

Country dependent citizen identifier. Document is sent as part of the customer object only for India to maintain customer's GST ID.

documentType	string

Details for *documentType*

Identifies the type of the data in the document field. Document type is sent as part of the customer object only for India to identify customer's document. By default, the document type is GST ID for India.

payment fields

paymentMethod	string	required
Details for paymentMethod Customer method of payment. Debit Card is not supported for country code BR. This value is obtained from the GET paymentmethods API. Typical values are VISA, MASTERCARD, AMERICAN_EXPRESS, DISCOVER, RUPAY, MAESTRO, CREDIT_CARD, DEBIT_CARD.		
countryCode	string of length 2 to 2	required
Details for *countryCode* ISO 3166-1 alpha-2 customer country code. To ensure proper conversion rates this value must be accurate.		
paymentType	string *(see details)*	
Details for *paymentType* The type of payment method used. Typically, not set directly as it is inferred from paymentMethod. • Type Values: CREDIT_CARD, DEBIT_CARD, PREPAID_CARD, BANK_TRANSFER, VIRTUAL_ACCOUNT, DIRECT_DEBIT, CASH, CHECK, OFFER, WALLET, CARRIER_BILLING		
storedPaymentProfileId	string	
Details for *storedPaymentProfileId* Supply the ID for a stored payment profile instead of providing explicit customer details.		

firstName	string of length 0 to 50
Details for firstName First name of customer. Some instruments use the full name passed as accountHolder.	
lastName	string of length 0 to 50
Details for lastName Last name of customer. Some instruments use the full name passed as accountHolder.	
accountHolder	string of length 0 to 100
Details for *accountHolder* Full name of customer. Some payments use the full name instead of firstName and lastName.	
streetAddress	string of length 0 to 100
streetAddress2	string of length 0 to 20
city	string of length 0 to 30
stateProvince	string of length 0 to 20
Details for *stateProvince* Customer state or province abbreviation (or full name in some countries)	
postalCode	string of length 0 to 20
Details for *postalCode* Customer postal code	
dateOfBirth	string of length 0 to 10

emailAddress	string of length 0 to 100
phoneNumber	string
Details for *phoneNumber* Customer mobile number frequently used to receive one time passcode	
document	string
Details for document Country dependent citizen identifier such as • CPF/CNPJ in BR • DNI/CUIT/CUIL in AR and PE • CURP in MX • RUT in CL • GST in IN This value is masked and an empty string is returned on retrieval.	
documentType	string
Details for *documentType* Identifies the type of the data in the document field. Typical values are from DNI, CI, CPF, CNPJ, RUT, CURP, GST	
cardNumber	string
Details for *cardNumber* Customer credit card number On retrieval most digits are masked with X's; only the last four digits are returned.	
expirationDate	string

Details for expirationDate Expiration Date of credit card in YYYY-MM format This value is masked and an empty string is returned on retrieval.	
securityCode	string
Details for *securityCode* Security Code or CVV of customer credit card This value is never persisted and an empty string is returned on retrieval.	
bankCode	string
Details for *bankCode* The customer method of payment for some paymentType BANK_TRANSFER instruments. Typically mirrors the paymentMethod field.	
bankNumber	string
channel	string
Details for *channel* The customer method of payment for some paymentType CASH instruments. Typically mirrors the paymentMethod field.	
password	string
Details for password Customer authentication secret such as a OTP This value is masked and an empty string is returned on retrieval.	
pin	string

Details for *pin* Customer authentication PIN This value is masked and an empty string is returned on retrieval.	
metadata fields	
instrumentConfigId	integer
Details for *instrumentConfigId* Used to declare which instrument was used when building the payment field values. It should be copied from the GET paymentmethods response body field paymentMethods[].instrumentConfigId. In general EP recommends merchants do not provide a value for this field.	
returnURL	string
Details for *returnURL* Merchant success return page URL. This is required in some cases such as when 3D-Secure is enabled. For more information refer to the paymentOption fields in the charge response.	
failureURL	string
Details for *failureURL* Merchant failure return page URL. This is required in some cases such as when 3D-Secure is enabled.	
cancelURL	string
Details for *cancelURL*	

Merchant cancel return page URL. This is required in some cases such as online banking. The cancelURL can be omitted for 3D-Secure.	
device	string *(see details)*
Details for *device* The display form factor of the device. When provided is used to optimize third party pages such as 3D-Secure. • Type Values: PC, TABLET, SMARTPHONE	
userAgent	string
Details for *userAgent* The HTTP User-Agent header from the customer's device. When provided is used to optimize third party pages such as 3D-Secure.	
locale	string *(see details)*
Details for *locale* This is a 2-letter ISO language code and 2-letter ISO country code separated by underscore. Typically, the locale is inferred from the customer's country. When provided is used to optimize third party pages such as 3D-Secure. • Type Values: en_US, fr_FR, pt_BR, ar_EG, da_DK, de_DE, es_ES, fi_FI, it_IT, ja_JP, ko_KR, nl_NL, no_NO, pl_PL, ru_RU, sv_SE, tr_TR	
pspOptions	string
Details for pspOptions	

An opaque string passed to the psp for custom integrations.		
InvoiceResponse fields		
id	string	required
transactionId	string	required
Details for *transactionId* Unique transaction ID generated by the payment system for original charge.		
transactionType	string	required
Details for *transactionType* States for which tax invoice was generated. Valid values are CHARGE,REFUND		
status	string	required
Details for status Status of the pdf generation. Valid values are INITIAL, COMPLETE		

Api authentication:

1. Introduction:

The Payzone payment platform validates API requests with a HMAC signature. Only requests with the correct signature are allowed to access the system resources.
The signature together with other important security information is expected to be sent in the request headers. You will receive an error message with HTTP status 401 if the signature is missing or invalid.
The document is to explain how to generate the signature.

2. API Caller Credentials:

You will receive these credentials when your merchant account is created. If you don't have these credentials, please contact your account manager.

Name	Description
Merchant Account Name	Assigned by Payzone. The unique merchant account name
Caller Name	Assigned by Payzone. The api caller ID
Caller Password	Assigned by Payzone. The api caller password, it is also the encrypted secret for HMAC signature.

3. Request Security Headers:

These HTTP request header values are required for each API call to the Payzone platform. Without any of these headers, the request is treated as an unauthorised request.

Header Name	Description
X-MerchantAccount	The Merchant Account name assigned to you.
X-CallerName	The name of the API Caller making the request
X-HMAC-Timestamp	Unix timestamp (seconds from epoch). Signatures generated with timestamps more than 30 minutes older than the request (as well as future timestamps) are rejected. e.g
X-HMAC-Signature	The generated signature

4. Generate the Signature:

Parameter definitions:

- **X-CallerName** - the name of the caller
- **X-MerchantAccount** - the name of the merchant account
- **X-HMAC-Timestamp** - current Unix time in seconds. This MUST be in UTC
- **request_path** - relative path of API, include query params if present

- **request_content** - The HTTP method body if present. Put it as an empty string (“”) if it is not present.

The signature can be generated as follows (pseudo-code):

```
var timestamp = seconds_from_epoch_utc
var message = string_concat(callerName, merchantAccountName, timestamp, request_path,
request_body)
var signature = hmac_sha256_as_hexadecimal(api_secret, message)
```

As a concrete example, suppose we have a caller calling the healthcheck endpoint:

- **X-CallerName** = “$caller”
- **X-MerchantAccount** = “MYNAME”
- **X-HMAC-Timestamp** = “1633767872”
- **Caller’s password** = “123456”
- **Request url** = “https://payment-sandbox.payzone.ma/api/v3/healthcheck”

It goes:
message = “$callerMYNAME1633767872/api/v3/healthcheck”
signature =
“B6693ABCCB887DD65B8DD05FAC5AC19653154C63006896ED4912EAAEBF10FEB1”

Finally, the request goes as below (in CURL):

```
curl --location --request GET 'https:// payment-sandbox.payzone.ma/api/v3/healthcheck' \
--header 'X-MerchantAccount: MYNAME' \
--header 'X-CallerName: $caller' \
--header 'X-HMAC-Timestamp: 1633767872' \
--header 'X-HMAC-Signature: B6693ABCCB887DD65B8DD05FAC5AC19653154C63006896ED4912EAAEBF10FEB1' \
--header 'Content-Type: application/json'
```

You will receive a HTTP 200 status with an empty response if your signature is correct.
You will see this error message when the signature is wrong:

```
{"requestId":"44d123b4-5cdc-4522-908b-
3d5765addbf3","errorCode":"authentication_error","message":"HMAC Authentication
failed. Invalid name or password"}
```

Get Transaction

The get transaction API helps to retrieve a charge or credit transaction detail. The API requires HMAC signature, please refer to *API Authentication* for more information about the signature.

API Endpoint	https://{{server}} /api/v3/charges/{id}
Method	GET
Content Type	Application-json

1. Overview

Field	Required	Description
id	Y	The transaction ID specified by the merchant.

2. Sample Response


```
{
   "id": "202110112305",
   "internalId": "2347897968234687234",
   "status": "CHARGED",
   "merchantAccount": "ZARA_test",
   "lineItem": {
       "amount": 12,
       "currency": "MAD",
       "description": "Revert 5 Golden Rings",
       "taxCode": "NT",
       "taxIncludedInPrice": true
   },
   "transactions": [
       {
           "timestamp": "2021-10-11T15:07:17Z",
           "gatewayProvidedId": "dd317ae1-f9cb-47ef-8a8b-10e8a754c5a2",
           "type": "CHARGE",
           "state": "APPROVED",
           "amount": 12,
           "currency": "MAD",
```

```
            "resultCode": 0,
            "responseText": "Success"
        }
    ],
    "paymentType": "CREDIT_CARD",
    "paymentMethod": "Visa",
    "created": "2021-10-11T15:07:16Z",
    "amountRemaining": 12
}
```

3. Sample Error Response

```
{
    "requestId": "f12fc754-5333-4a51-9c27-ef443f4f8e89",
    "errorCode": "entity_not_found_error",
    "message": "Entity not found: Cannot find charge with ID: 202110111222"
}
```

Capture:

This method updates an existing Charge by issuing a new command to it.

API Endpoint	https://{{server}}/api/v3/charges/{id}
Method	POST
Content Type	Application-json

1. Request Path Parameters:

Field	Required	Description
id	Y	The transaction ID specified by the merchant.

2. Request Parameters:

Field	Required	Description
command	Y	• SETTLE completes a charge initiated with AUTHORIZE

3. Sample Capture Request:

```
{
 "command": "SETTLE",
"amount": 12,7
}
```

4. Sample Capture Response

Check the last transactions[n] item, if transactions[2].type = “SETTLE”
and transactions[2].state = “APPROVED”


```
{
   "type": "v3_2",
   "id": "202110112335",
   "status": "CHARGED",
   "merchantAccount": "DevTest",
   "lineItem": {
       "amount": 12.7,
       "currency": "MAD",
       "description": "5 Golden Rings",
       "taxCode": "NT",
       "taxIncludedInPrice": true
   },
   "transactions": [
       {
           "timestamp": "2021-10-11T15:35:58Z",
           "gatewayProvidedId": "2fe1bf49-ce48-43be-8c4d-b22a1f364c19",
           "type": "AUTHORIZE",
           "state": "APPROVED",
           "amount": 12.7,
           "currency": "ZAR",
           "resultCode": 0,
           "responseText": "Success"
```

```
        },
        {
            "timestamp": "2021-10-11T15:36:51Z",
            "gatewayProvidedId": "61100cb2-fd58-45d2-85c8-699773c951e0",
            "type": "SETTLE",
            "state": "APPROVED",
            "amount": 12.7,
            "currency": "MAD",
            "resultCode": 0,
            "responseText": "Success"
        }
    ],
    "paymentType": "CREDIT_CARD",
    "paymentMethod": "VISA",
    "created": "2021-10-11T15:35:56Z"
}
```


Cancel:

This method updates an existing Charge by issuing a new command to it.

API Endpoint	https://{{server}}/api/v3/charges/{id}
Method	POST
Content Type	Application-json

1. Request Path Parameters:

Field	Required	Description
id	Y	The transaction ID specified by the merchant.

2. Request Parameters:

Field	Required	Description
command	Y	• AUTH_REVERSAL cancels an AUTHORIZE and releases the reserved funds on the customer's credit card

3. Sample Canceled Request

```
{
 "command": "AUTH_REVERSAL",
"amount": 12,7
}
```

4. Sample Canceled Response

Check the last transactions[n] item, if transactions[2].type = “AUTH_REVERSAL” and transactions[2].state = “APPROVED”


```
{
   "type": "v3_2",
   "id": "202110112335",
   "status": "CHARGED",
   "merchantAccount": "DevTest",
   "lineItem": {
       "amount": 12.7,
       "currency": "MAD",
       "description": "5 Golden Rings",
       "taxCode": "NT",
       "taxIncludedInPrice": true
   },
   "transactions": [
       {
           "timestamp": "2021-10-11T15:35:58Z",
           "gatewayProvidedId": "2fe1bf49-ce48-43be-8c4d-b22a1f364c19",
           "type": "AUTHORIZE",
           "state": "APPROVED",
           "amount": 12.7,
           "currency": "MAD",
```

```
            "resultCode": 0,
            "responseText": "Success"
        },
        {
            "timestamp": "2021-10-11T15:36:51Z",
            "gatewayProvidedId": "61100cb2-fd58-45d2-85c8-699773c951e0",
            "type": " AUTH_REVERSAL ",
            "state": "APPROVED",
            "amount": 12.7,
            "currency": "MAD",
            "resultCode": 0,
            "responseText": "Success"
        }
    ],
    "paymentType": "CREDIT_CARD",
    "paymentMethod": "VISA",
    "created": "2021-10-11T15:35:56Z"
}
```


In case of SETTLE failure, the authorized amount is released to the payer.

Refund:

This method updates an existing Charge by issuing a new command to it.

API Endpoint	https://{{server}}/api/v3/charges/{id}
Method	POST
Content Type	Application-json

3. Request Path Parameters:

Field	Required	Description
id	Y	The transaction ID specified by the merchant.

4. Request Parameters:

Field	Required	Description
command	Y	• REFUND cancels an AUTHORIZE and releases the reserved funds on the customer's credit card

5. Sample Refund Request

```
{
 "command": "REFUND",
"amount": 12,7
}
```

6. Sample Refund Response

Check the last transactions[n] item, if transactions[2].type = “REFUND”
and transactions[2].state = “APPROVED”


```
{
   "type": "v3_2",
   "id": "202110112335",
   "status": "CHARGED",
   "merchantAccount": "DevTest",
   "lineItem": {
       "amount": 12.7,
       "currency": "MAD",
       "description": "5 Golden Rings",
       "taxCode": "NT",
       "taxIncludedInPrice": true
   },
   "transactions": [
       {
           "timestamp": "2021-10-11T15:35:58Z",
           "gatewayProvidedId": "2fe1bf49-ce48-43be-8c4d-b22a1f364c19",
           "type": "AUTHORIZE",
           "state": "APPROVED",
           "amount": 12.7,
           "currency": "MAD",
           "resultCode": 0,
```

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```
            "responseText": "Success"
        },
        {
            "timestamp": "2021-10-11T15:36:51Z",
            "gatewayProvidedId": "61100cb2-fd58-45d2-85c8-699773c951e0",
            "type": " REFUND ",
            "state": "APPROVED",
            "amount": 12.7,
            "currency": "MAD",
            "resultCode": 0,
            "responseText": "Success"
        }
    ],
    "paymentType": "CREDIT_CARD",
    "paymentMethod": "VISA",
    "created": "2021-10-11T15:35:56Z"
}
```